Accelerators producing high energy He beams can also be used because the high energy α-particles will penetrate through the vial walls. Then there will be no residual activity in the sample. Recently a new method based on "in vitro" radiation of solution using $^{211}$At has been developed at Chalmers. This method will also produce an, in principle, non radioactive solution with radiolysis products.

Much insight into radiation processes and radiation induced radical reactions has been gained by means of the *pulse radiolysis* method which is based on irradiation of samples with a short pulse of ionizing radiation. The radiation source is generally an electron accelerator or, to a much lesser extent, a positive particle accelerator. Techniques used to follow the transient behaviour of the radiolytically generated short-lived species are optical absorption spectroscopy, esr, conductivity and polarography. The time resolution is generally in the femtosecond to microsecond range.

## 8.10. DOSE MEASUREMENTS

The amount of radiation energy absorbed in a substance is measured with *dose meters* (or *dosimeters*). These may react via a variety of processes involving (a) the heat evolved in a calorimeter. (b) the number of ions formed in a gas, (c) the chemical changes in a liquid or in a photographic emulsion, and (d) the excitation of atoms in a glass or crystal. The first two ones are *primary meters* in the sense that they can be used to accurately calculate the exposure or dose absorbed from a radiation source. They can be used to calibrate the *secondary meters*.

In 1925 C. D. Ellis and W. A. Woorter, using RaE ($^{210}$Bi) in *calorimetric measurements*, obtained the first proof that the maximum energy and the average energy of β-radiations were different. A precision of about 1% can be obtained in a calorimeter for an energy production rate of $\sim 10^{-6}$ J/s which corresponds to approximately 0.7 MeV average energy for a sample of 40 MBq (1 mCi). If the average energy of an α- or β-emitting nuclide is known, calorimetric measurement of the energy production rate can be used to calculate the specific activity. This technique is not suitable for γ-sources.

A more sensitive and general instrument for the measurement of ionizing radiation is the *condenser ion chamber.* This is a detector which has a small gas-filled volume between two charged electrodes. When radiation ionizes the gas between the two electrodes, the cations travel to the cathode and the electrons to the anode, thus preventing recombination of the ion pairs. Measurement of the amount of discharge provides a determination of the ionization and consequently of the dose delivered to the instrument. This type of instrument is described in more detail in Ch. 9. The flexibility and accuracy of this dosimeter have led to it being widely employed for the exact measurement of γ-dose rates. The most common version of this type of instrument is the pen dosimeter (Fig. 8.8), which can be made to provide either a direct reading of the absorbed dose or

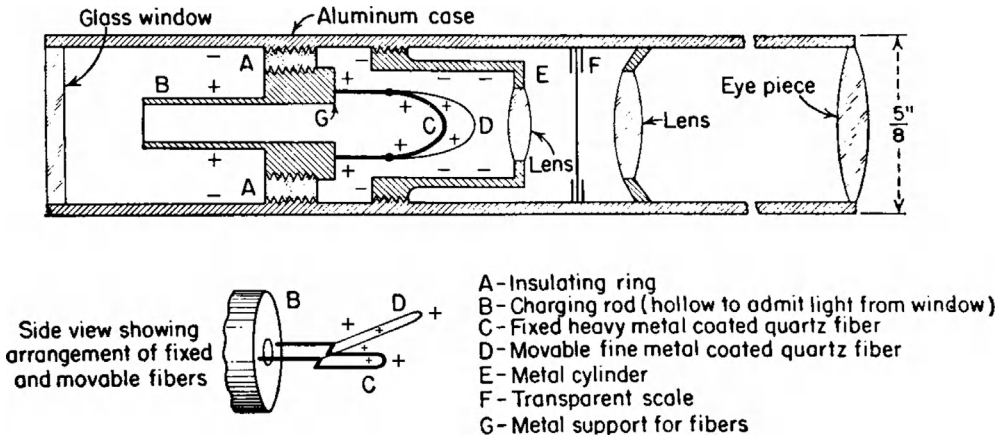



**Figure 8.8** Pen dosimeter with direct readout of the dose. *(From Lapp and Andrews, 1948.)*

indirectly via an auxiliary reading instrument. Instruments with ranges from 0.0002 to 10 Gy (full scale) are available commercially.

There are numerous *chemical dosimeters* based on the radiolysis of chemical compounds, both organic and inorganic. An illustrative example is the $CHCl_3$ dosimeter. This is a two- phase aqueous-organic system. Radiation produces HCl which changes the pH of the almost neutral aqueous phase as shown by the colour change of a pH indicator. This dosimeter is suitable only for rather high doses, $10^2-10^5$ Gy.

The most common chemical dosimeter is the *Fricke dosimeter* (§8.7), which consists of an aqueous solution of approximately the following composition: 0.001 M $Fe(NH_4)_2(SO_4)_2$, 0.001 M NaCl, and 0.4 M $H_2SO_4$. The amount of $Fe^{3+}$ formed through irradiation is determined spectrophotometrically and the dose absorbed in Gy calculated by the equation:

$$D(\text{Gy}) = A/\{\varepsilon\, x\, \rho\; G(Fe^{3+})\} \quad (\text{J/kg}) \tag{8.40}$$

where $A$ is the change in absorbance, $G(Fe^{3+})$ is the yield of $Fe^{3+}$ in mol/J (cf. eqn. 8.33), $\varepsilon$ is the molar extinction coefficient (217.4 $m^2$/mol at 304 nm), $x$ is the length of the cell (in m), and $\rho$ is the density of the solution (1024 kg/$m^3$ at 15–25°C). The $G$-value depends somewhat on the *LET* value of the radiation as seen in Figure 8.7. The Fricke dosimeter is independent of dose rate up to dose values of about $2 \times 10^6$ Gy/s and can be used in the range of 1–500 Gy. In a common modification, the solution also contains NaSCN, leading to the formation of the intensely red complex ion $Fe(SCN)_6^{3-}$ upon irradiation.

*Photographic emulsions* are sensitized by ionizing radiation resulting in darkening upon development. This is used in the film dosimeter for measurement of β-, γ-, or n-doses. In order to differentiate between various types of radiation, the film can be surrounded

by filters or transfer screens. Although any type of film may be used, special nuclear emulsions have been designed. The dose received is directly proportional to the optical density of the exposed film. Film dosimeters are useful in the same range as the pen dosimeter, and both are used for personnel measurements. While the pen dosimeter can be read directly, the film dosimeter requires development.

The *glass dosimeter* is made of phosphate glass containing 5−10% of $(AgPO_3)_n$ polymer. The small piece of glass, a few cubic centimeters in size, is protected from light by means of a coating. When radiation strikes the glass, trapped electrons are produced which can be released by irradiation with ultraviolet light after removal of the protective coating. This results in the emission of fluorescent radiation which can be measured photometrically. The amount of fluorescent radiation is proportional to the dose received for doses up to 10 Gy.

The *thermoluminescent dosimeter* (*TLD*) covers the range $10^{-6}$−10 Gy. The detector consists of a crystalline powder of $CaF_2$, LiF, or similar compound, either pure or incorporated in a plastic material like teflon. The irradiation leads to ionization and trapping of the electrons in imperfections in the crystal lattice. Upon heating, recombination occurs with light emission, which is measured photometrically. The electrons are trapped at different energy levels, and slow heating releases the electrons in order of increasing energy of the trapping levels. Consequently it is possible to take a dose reading on the lowest-energy trapped electrons and still retain a memory of the dose through the electrons left in more energetic traps. The dose can be read at a later time by releasing the remaining electrons at a higher temperature. The TLD can be designed like the pen dosimeter in which it is surrounded by screens to differentiate between different kinds of radiation. By using a lead filter the dosimeter can be made energy independent in the range 0.02−20 MeV for X-rays and $\gamma$-radiation.

This is also the basis for *thermoluminescence dating*. When geologic minerals like quartz, feldspars, etc as well as ceramic materials like fired clay are exposed to high energy cosmic radiation during the ages, imperfections are produced in the crystal lattice. When heated these crystals produce light in proportion to the dose received. If the dose rate is known the age of the exposed material can be calculated. There are numerous examples of the use of this technique e.g. for authentication of old porcelain or ceramics. Recently British Museum was forced to remove a large number of Mexican, Greek and Roman sculptures and vases because TLD-dating showed that they had been manufactured during the 19th century. Also samples of the enamel covering human teeth can be used to determine the accumulated dose by thermoluminescense. However, irradiation of the teeth by UV-light, often used by dentists, will erase some − or all − of the accumulated energy.

A modification is the *thermocurrent dosimeter*. In this case the detector may be a thin crystal of synthetic sapphire ($\phi$ 10 mm, thickness $\leq$1 mm) between thin metal electrodes. When the crystal is heated after having received a certain dose, the trapped electrons are released and cause a current to pass between the electrodes. The peak of the integrated

current is a measure of absorbed dose. The effect is referred to as RITAC (radiation induced thermally activated current) and the technique is named TC (thermocurrent) dosimetry. Because it is instrumentally easier to measure an electric current than light, the TC dosimeter may replace the TLD with time.

Doses can also be calculated from the product of the dose rate and the time of exposure. The most common *dose rate meter* is the ionization chamber. Because of the close connection between this instrument and pulse type ionization counters, which measure individual nuclear particles entering the detector, the discussion of ionization chambers is deferred to Ch. 9, which deals in more detail with radiation measurement techniques.

If the number, energy, and type of nuclear particles being absorbed in a material can be measured or estimated, the absorbed dose can be calculated as described previously. Such calculations are very important, particularly in the field of radiation protection.

## 8.11. LARGE-SCALE NON-BIOLOGICAL APPLICATIONS

In this section we deal only with the more important non-biological applications, as the latter are treated in Ch. 15, which specifically deals with biological effects of radiation.

Ionizing radiation produces ionized and excited atoms and molecules in all materials. Excited molecules formed directly or by recombination reactions between electrons and cations decompose in the vast majority of systems to highly reactive free radicals. The reactive species formed on radiolysis are precursors of further reactions, such as reduction, oxidation, polymerization, cross linking and so on. It should therefore be possible to apply radiation chemical methods to industrial processes and, consequently, extensive applied research and development of radiation chemistry has been carried out during the past three decades.

A large proportion of the radiation induced reactions can be brought about by thermal, photochemical or chemical initiation, but the advantages of radiation initiation are generally claimed to be:

- No contamination by catalyst and catalyst residue.
- Temperature independence.
- Easy control of radiation intensity and hence rate of induced reactions.
- High speed treatment capability.
- Ionizing radiation offers the advantage of greater penetrating power compared to initiation by UV-light.

### 8.11.1. Radiation sources

The radiation sources normally used can be divided into two groups, those employing radioactive isotopes (e.g. $^{60}$Co) and those employing electron beam accelerators (EBA).